ABHISHEK DASS

Flight Software Developer & Software Engineer | Coding Evaluation • Data Annotation & Labeling • Model Output Review

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PROFILE

Computer Science graduate (June 2026) with hands-on experience across full-stack software development, embedded/flight software, ETL data pipelines, machine-learning anomaly detection, and real-time systems evaluation. Currently a Flight Software Developer on SUTS, a CubeSat mission by the Estonian Student Satellite Foundation, building mission-critical satellite software with an international student team — work that demands rigorous correctness checks, structured testing, and careful review of every submitted change. Comfortable designing structured coding problems, building test/verification logic, cleaning and validating large datasets, and assessing output correctness and quality — core skills for remote, task-based AI training and evaluation work.

FIT FOR AI TRAINING & EVALUATION WORK

- **Coding-task design & evaluation:** write and review mission-critical flight software (Estonian Student Satellite) and backend/API code (Java, Python, JavaScript, C/C++) — able to author self-contained coding problems, reference solutions, and test/verifier logic.
- **Data labeling & annotation:** built a production web scraper (SaafAI) plus ETL pipelines processing 10,000+ records, with data extraction, normalization, and quality checks across MongoDB, MySQL, and PostgreSQL.
- **Response ranking / output evaluation:** built anomaly-detection and monitoring systems (CNN-LSTM, real-time telemetry) that flag incorrect or anomalous outputs against expected behavior — directly analogous to judging model output quality.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, Python, SQL, C, C++, C#

Backend & Web Development: Node.js, Express.js, Spring Boot, REST APIs, JWT Authentication, .NET, Back-End Web Development, Web Development, Web Scraping

Frontend Development: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, SQL

Data Engineering & Analytics: Data Analysis, Data Engineering, Data Extraction, Data Entry, Data Collection, Data Cleaning, Data Acquisition, Data Normalization & Transformation, ETL Pipelines, Qualitative Data Analysis, Qualitative Analysis, Pandas, Dashboarding & Reporting, Power BI, Tableau, Microsoft Excel

Machine Learning & AI: Machine Learning, Machine Learning Algorithms, CNN-LSTM, Anomaly Detection, Object-Oriented Programming (OOP)

Embedded, Systems & Engineering Design: Flight/Embedded Software, NI LabVIEW, MQTT, SolidWorks, Fusion 360, Ansys, 3D Design, Aerodynamics

Tools & Platforms: Git, GitHub, Postman, Ubuntu, Microsoft 365 Copilot

Core CS Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, System Design, Cybersecurity

RELEVANT EXPERIENCE

Flight Software Developer — Estonian Student Satellite Foundation — SUTS CubeSat Mission (Launch 2028) *Jul 2026 – Present*

- Developing flight software for mission-critical satellite systems as part of an international student engineering team, where code must pass strict correctness and review standards before deployment.
- Collaborating asynchronously across time zones — writing, documenting, and verifying software to shared mission specifications, directly mirroring remote, async task-based evaluation work.

LabVIEW Intern — Hixaa Technologies Private Limited *Jan 2026 – Jul 2026*

- Developed LabVIEW-based applications for a hearing-focused medical instrument, a metro rail simulator, and product software-licensing systems.
- Built and tested instrument/simulator logic against defined behavioral specs — verifying outputs matched expected results before release.

Junior Software Engineer — PugArch Technology Pvt. Ltd. *Jul 2025 – Feb 2026*

- Built SaafAI, a production web scraper for extracting targeted data at scale to populate a structured database — direct experience with large-scale data collection and cleaning.
- Developed a map integrated with the scraped database, and built Python ETL pipelines processing 10,000+ structured records across MongoDB, MySQL, and PostgreSQL.
- Reduced manual preprocessing effort by 60% by building automated data-quality checks.

Data Analytics Intern — Infosys *Aug 2025 – Oct 2025*

- Analyzed datasets using SQL, Excel, and Power BI; built interactive dashboards and performed systematic data cleaning and quality checks to ensure reporting accuracy.
- Reduced reporting turnaround time by 40% through workflow automation.

Aerodynamics & Data Acquisition Head — E-Formula AshwaRiders (Formula Student Team) *May 2023 – Feb 2026*

- Progressed from Jr. Aerodynamic Design Engineer (SolidWorks, Fusion 360, Ansys) to Data Acquisition Head to Aerodynamics & Data Acquisition Head, also serving as Inventory Manager — a 2 years 10 months tenure across engineering, data, and operations.
- Designed and built real-time vehicle telemetry pipelines (LabVIEW, MQTT), with monitoring dashboards that compared live outputs against expected performance ranges to catch anomalies during testing.
- Led aerodynamics and data-acquisition strategy contributing to an AIR 8 overall finish at Formula Bharat 2024 against 40+ competing teams.

KEY PROJECTS

CivicTrack – Public Issue Reporting Platform — *Java, Spring Boot, SQL, REST APIs*

- Designed and built REST APIs for issue creation, status tracking, and department allocation — self-contained, testable problem specs similar to structured coding-evaluation tasks.

DL-SEC – Cybersecurity Anomaly Detection System (co-authored research paper) — *Python, Machine Learning, CNN-LSTM*

- Built and co-authored a research paper on a CNN-LSTM anomaly-detection pipeline that scores streaming data and flags deviations from expected patterns — the same judge-the-output logic used in AI response evaluation.

Real-Time Vehicle Telemetry & Digital Analytics System — *Python, LabVIEW, PostgreSQL, MQTT, ETL*

- Built an ETL pipeline processing 1,000+ sensor records per session, with automated checks against expected value ranges and dashboards for anomaly detection.

CrisisNet – Disaster Response Coordination Platform — *React.js, Node.js, MongoDB, Socket.IO*

- Built a real-time, full-stack coordination platform with live location tracking and request-routing logic for emergency response teams.

EDUCATION

B.Tech / B.E. – Computer Science and Business Systems *Expected Graduation: June 2026*

St. Vincent Pallotti College of Engineering and Technology, Nagpur | CGPA: 8.69 / 10

CERTIFICATIONS

- Data Analytics Internship Certification – Infosys
- MERN Stack Web Development – Edunet
- AI-ML and Green Technology – Edunet
- Data Structures and Algorithms in Java
- Programming Foundations: Fundamentals; Programming Foundations: Beyond the Fundamentals
- Artificial Intelligence in Marketing; AI Tools Workshop; Learning Microsoft 365 Copilot

ACHIEVEMENTS & LEADERSHIP

- Selected as Flight Software Developer for SUTS, a CubeSat mission by the Estonian Student Satellite Foundation (launch 2028), joining an international student engineering team.
- AIR 8 overall finish at Formula Bharat 2024 (against 40+ competing teams) as Aerodynamics & Data Acquisition Head.
- Co-authored a research paper on CNN-LSTM based anomaly detection (DL-SEC).
- Built production-oriented telemetry and evaluation systems used for live engineering decisions.